

ARUN SHARMA

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SUMMARY

I build physics-informed AI systems that make spatial reasoning reliable for real-world deployment, from detecting GPS spoofing to enable safer autonomous navigation to modeling energy-efficient vehicle routing to downscaling climate data for coastal flood risk assessment. My research fuses deep generative models (diffusion models, transformers), physics-informed learning, and geostatistical priors to address data incompleteness, distribution shift, and violations of physical laws in spatial ML. CS PhD from UMN (2026). At Esri, I shipped a maritime anomaly detection pipeline on AWS that improved classification accuracy from 55% to 73% and reduced API latency by 30%. Seeking full-time research opportunities.

EDUCATION

University of Minnesota, Twin Cities – Ph.D., Computer Science

2021 – 2026

Advisors: Prof. Shashi Shekhar | **Committee:** Prof. Vipin Kumar, Prof. Ravi Janardan, and Prof. Ying Song

Dissertation: Distortion-Aware Spatial Data Science | **Doctoral Dissertation Fellowship (2022–23)**

University of Minnesota, Twin Cities – M.S., Computer Science

2018 – 2021

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, Scala, C/C++, Julia, Rust, Go, Shell (Bash), MATLAB, JavaScript (Node.js), TypeScript.

Machine Learning: MLflow, XGBoost, MLlib, LLMs (fine-tuning & inference), VLMs, RAG, LangChain, VectorDB (FAISS, Chroma, Pinecone), Hugging Face Transformers/TRL, LoRA, PEFT, RLHF, OpenAI API, Triton Inference Server, ONNX.

Agentic AI: Chain-of-Thought (CoT), ReAct, Tool/Function Calling, Agent Traceability, Tool-Call & Token Optimization, Multi-Agent Orchestration (LangGraph, AutoGen), MCP, Prompt Caching.

Big Data & Distributed Systems: PySpark, Flink, Presto, Airflow, Kafka, RabbitMQ, Delta Lake, Docker, Terraform, Kubernetes.

ML Frameworks & Deployments: PyTorch, TensorFlow, JAX, ONNX, SageMaker, Lambda, ECS, Bedrock, vLLM/SGLang, Temporal-Nexus.

RL & Alignment: RLHF/RLAIF/RLVR, PPO, DPO, GRPO, Constrained RL, Reward Hacking Mitigation, Agent Traceability, Tool-use Safety.

EXPERIENCE

ESRI – Environmental Systems Research Institute

Research Scientist Intern, May 2023 – Dec 2023

- Improved detection of illegal vessel behavior (route deviations, dark shipping) from 55% to 73% accuracy by independently designing and deploying an end-to-end anomaly detection pipeline integrating Transformer-based models, Evidential Deep Learning (EDL), and AWS SageMaker on a large-scale maritime dataset (~500M AIS records), with automated retraining via Lambda, ECS, and Step Functions.
- Reduced maritime route query latency by 40% to support real-time vessel tracking by designing and implementing a scalable Graph-based Traffic Representation and Association (GTRA) framework leveraging PySpark and GeoAnalytics APIs over a large-scale AIS maritime dataset.
- Cut model retraining time by 35% and API latency by 30% to enable faster anomaly alerting by implementing model quantization and optimizing inference with SageMaker Multi-Model Endpoints (MME), Step Functions, SQS, CloudWatch, and CI/CD pipelines.

University of Minnesota, Twin Cities

Graduate Research Assistant Aug 2018 – Aug 2025

- Led Pi-DPM, a physics-informed diffusion model flagging GPS-spoofed and AI deep-fake trajectories for maritime and autonomous-vehicle security: a transformer Context-Informed Encoder (spatial and temporal attention over neighbor trajectories) plus a physics-informed decoder with a single-axle Kinematic Bicycle Model (S-KBM) prior and velocity, acceleration, heading, curvature, and turning-rate regularizers. Beat 6 state-of-the-art baselines (DiffWave, DiffTraj, ControlTraj) at 5/10/20% anomaly rates with strong transfer within and across maritime and urban domains.
- Co-led a cross-disciplinary team to sharpen coastal flood-risk assessment, designing a Kriging-informed Conditional Diffusion Probabilistic Model that downscales coarse climate projections (25km to 1km) into fine sea-level maps: a U-Net diffusion conditioned on Universal Kriging interpolations under a Matern variogram, with a regularizer penalizing the gap between empirical and observed variograms during reverse diffusion. Using Copernicus CDS and CMIP6 HighResMIP (EC-Earth3P) data, it produced high-resolution sea-level-anomaly and eddy-kinetic-energy maps and outperformed bicubic, CNN, GAN, and baseline diffusion models on RMSE, MAE, PCC, and CRPS across eastern and western North America.
- Designed Pi-GRPO, a physics-informed RL stack for trajectory generation and reasoning policies that unifies PPO, DPO, and GRPO under one hybrid reward (unbounded hard S-KBM violation penalty, 95th-percentile jerk/curvature soft envelope, Pi-DPM reconstruction-likelihood term, optional cross-encoder preference model), with GRPO group-baseline advantages (no value head), physics-aware gamma_phys DPO, and PPO under bounded adaptive KL, served by vLLM rollouts with prefix caching, content-addressed checkpoints, and W&B/OpenTelemetry reward decomposition. Trained on ~11K preference triples curated from GeoTrace-Agent human review, it reduced the hard-violation rate from 18% (vanilla DPO) to 0% (physics-augmented DPO) while keeping KL bounded over 3,000 PPO steps.
- Built GeoTrace-Agent, a production-grade multi-agent framework for auditable spatiotemporal reasoning over heterogeneous trajectory and Earth-observation sources: a typed PlanGraph chain-of-thought planner decomposes natural-language queries into deterministic sub-tasks across AIS, OSM road networks, Copernicus weather, Sentinel imagery, and space-time-prism tools, routed to specialized agents (Hagerstrand prism reasoning, STAGD-DRM gap detection, TGARD/DC-TGARD rendezvous discovery, S-KBM kinematic validation). Exposed over MCP and JSON-RPC A2A with OpenTelemetry tracing, Postgres human-in-the-loop queues, semantic caching, tool-call deduplication, and prompt compression; cut per-query token spend ~40% (about \$0.054 to \$0.034 per query) and blocked physically infeasible regions before user output.
- Co-designed GCDM (Geo-lucid Conditional Diffusion Model), a two-stage cascaded diffusion generating realistic synthetic vehicle trajectories for autonomous-driving simulation and traffic prediction by injecting attributed road-network priors via map-informed latent variables and spatial cross-attention, with a physics-based decomposition (map-derived base plus learned residual). Beat 4 baselines (DiffTraj, Sashimi, DiffWave, EETG) on

- Porto (about 1.7M trips) and Harbin taxi data on geo-distribution similarity (density, trip error) and velocity, acceleration, and jerk fidelity, and lowered downstream traffic in-flow and out-flow prediction error.
- Co-designed MBOR (Multi-Level Bi-Objective Routing) for real-time electric-vehicle route planning that balances energy use against travel time over fragmented road networks, introducing three novel data structures, a boundary multigraph, a Multi-level Encoded Pareto Frontier View (MEPFV), and 2D cost-interval pruning with multi-edge pruning, to precompute and retrieve complete Pareto-optimal path sets. Ran more than 10x faster online than prior methods on real-world road network data.
 - Led a multi-year, four-publication program (homeland security and public safety) that cut the area investigators must manually search for illegal maritime activity (illegal fishing, oil transfers, trans-shipments) by up to 80%. Built STAGD (Space-Time Aware Gap Detection) with a Dynamic Region Merge criterion combining comparison-less temporal indexing, R*-tree hierarchical spatial indexing, and maximal-union merge over space-time prism geo-ellipses, then TGARD and a Dual Convergence variant (DC-TGARD) exploiting ellipse symmetry via bi-directional pruning and early stopping; beat baselines on speed and accuracy over AIS (Automatic Identification System) data covering 500K ships.
 - Built SM-Hybrid, a novel surrogate for the computationally expensive Daycent agroecosystem model that uses hybrid spatial neural network architectures to capture both spatial autocorrelation (within-site dependencies) and teleconnections (cross-site long-range dependencies). Predicted land emissions (GPP, Ra, Rh, NEE, crop yield) across Iowa, Illinois, and Indiana are 2x faster than Daycent while beating SM-ANN baselines on accuracy, enabling rapid scenario evaluation for climate-smart agricultural policy.
 - Built domain-adapted Spatially-Lucid neural networks for precision immunotherapy, classifying immune-tumor cell relationships via multi-category point set classification in non-Euclidean space with a spatial ensemble framework (place-calibration parameters, weighted-distance learning rate, spatial domain adaptation across heterogeneous tissue place-types). Used a multi-task architecture with spatial mix-up masking and spatial contrastive predictive coding for self-supervised learning on MxIF (multiplexed immunofluorescence) oncology data.
 - Built a community-engaged spatial-epidemiology system measuring how COVID-19 policies shifted mobility: designed the entity-relationship model, system architecture, and implementation over aggregated mobile-device data, delivering county-level analytics on long-duration visits to high-risk business categories and fine-resolution device-count maps to epidemiologists, analysts, and policymakers. Collaborated directly with the Minnesota Department of Management and Budget; the work was cited in testimony to the MN House Transportation Finance Committee.

SELECTED PUBLICATIONS

- [1] Towards Physics-informed Diffusion for Anomaly Detection in Trajectories: A Summary of Results
 Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection (GeoAnomalies '25)
Arun Sharma, Mingzhou Yang, Majid Farhadloo, Subhankar Ghosh, Bharat Jayaprakash, and Shashi Shekhar
- [2] Towards Kriging-informed Conditional Diffusion for Regional Sea-Level Data Downscaling: A Summary of Results
 32nd International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24)
 Subhankar Ghosh*, **Arun Sharma***, Jayant Gupta, Aneesh Subramanian and Shashi Shekhar (*Both authors contributed equally to this paper)
- [3] Physics-based Abnormal Trajectory-Gap Detection
 ACM Transactions in Intelligent Systems and Technology, 2024.
Arun Sharma, Subhankar Ghosh, and Shashi Shekhar
- [4] Analyzing Trajectory Gaps for Possible Rendezvous Regions
 ACM Transactions in Intelligent Systems and Technology, 2022
Arun Sharma and Shashi Shekhar
- [5] Towards a Tighter Bound on Possible-Rendezvous Areas: Preliminary Results
 30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '22)
Arun Sharma, Jayant Gupta, Subhankar Ghosh, and Shashi Shekhar
- [6] Geo-lucid Conditional Diffusion Models for High Physical Fidelity Trajectory Generation
 Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)
 Mingzhou Yang, **Arun Sharma**, Majid Farhadloo, Bharat Jayaprakash, and Shashi Shekhar
- [7] Towards Surrogate Models with Hybrid Spatial Neural Networks: A Summary of Results
 Proceedings of the 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25), pp. 57–69, 2025
 Shengya Zhang, **Arun Sharma**, Majid Farhadloo, Mingzhou Yang, Yao Zhang, Mu Hong, Licheng Liu, David Mulla, and Shashi Shekhar
- [8] Towards Fine-Tuning-Based Site Calibration for Knowledge-Guided Machine Learning: A Summary of Results
 5th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)
 Ruolei Zeng, **Arun Sharma**, Shuai An, Mingzhou Yang, Shengya Zhang, Licheng Liu, David Mulla, and Shashi Shekhar
- [9] Spatially-Delineated Domain-Adapted AI Classification: An Application for Oncology Data
 SIAM International Conference on Data Mining (SIAM SDM) 2025
 Majid Farhadloo, **Arun Sharma**, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar
- [10] Towards Spatially-Lucid AI Classification in Non-Euclidean Space: An Application for MxIF Oncology Data
 SIAM International Conference on Data Mining (SIAM SDM) 2024
 Majid Farhadloo, **Arun Sharma**, Jayant Gupta, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar

REVIEWER: NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, IJCAI, SIGKDD, CIKM, ICDM, SDM, MLSys, SIGSPATIAL, TKDE, JMLR